

1 Pre-Requisites

1.1 Distributions

Binomial $X \sim \text{Bin}(n, p)$: $M(t) = (pe^t + 1 - p)^n$

Geometric $X \sim \text{Geometric}(p)$: $M(t) = \frac{pe^t}{1 - (1-p)e^t}$

Gamma $X \sim \Gamma(\alpha, \lambda)$:

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}.$$

Multinomial $X \sim \text{Multinomial}(n, \mathbf{p})$:

$$P(\mathbf{X} = \mathbf{x}) = \binom{n}{x_1, \dots, x_k} \prod_{i=1}^k p_i^{x_i}, \|\mathbf{x}\|_1 = n.$$

$$\text{Cov}(X_i, X_j) = -np_i p_j, \quad i, j \in [1, k], i \neq j.$$

Hypergeometric $X \sim \text{Hypergeometric}(N, m, n)$:

For $i \in [\max(0, n + m - N), \min(n, m)]$,

$$P(X = x) = \frac{\binom{m}{x} \binom{N-m}{n-x}}{\binom{N}{n}}.$$

$$E(X) = \frac{nm}{N}, \text{Var}(X) = \frac{nm}{N} \frac{N-m}{N} \frac{N-n}{N-1}.$$

1.2 Random Variables

Expectation and Variance

$$\text{Cov}(X, Y) = E\{(X - \mu_X)(Y - \mu_Y)\}.$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

The **moment generating function** is defined:

$$M_X(t) = E(e^{tX}) = \phi_X(e^t).$$

Properties of MGF:

- A MGF uniquely identifies a distribution

- $E[X^k] = \frac{d^k}{dt^k} M_X(t)|_{t=0}$

- $M_{aX+b}(t) = e^{bt} M_X(at)$

The **probability generating function** is defined for discrete RVs:

$$\phi_X(t) = E[t^X] = M_X(\ln t).$$

Properties of PGF:

- A PGF uniquely identifies a discrete RV's distribution

- $P(X = 0) = \phi_X(0)$

- $P(X = k) = \frac{1}{k!} \frac{d^k}{dx^k} \phi_X(t)|_{t=0}, \quad k \in \mathbb{N}^*$

Independence $\iff \forall x, y \quad f_{X|Y}(x|y) = f_X(x)$

$$\iff \forall s, t \quad M_{XY}(s, t) = M_X(s) M_Y(t)$$

$$\iff \forall s, t \quad \phi_{X+Y}(s, t) = \phi_X(s) \phi_Y(t)$$

$$\implies E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

$$\implies \text{Cov}(X, Y) = 0$$

Normal $X, Y \wedge \text{Cov}(X, Y) = 0 \implies \text{Independence}$

Conditional Distributions

We say $X_1, \dots, X_n$ are conditionally independent given Y if

$\forall \langle x_1, \dots, x_n, y \rangle \in \mathbb{R}^{n+1}$,

$$P(X_1 = x_1, \dots, X_n = x_n | Y = y)$$

$$= \prod_{i=1}^n P(X_i = x_i | Y = y)$$

$$p_{Y|X}(y|x) = \frac{p_{X|Y}(x|y) p_Y(y)}{\sum_y p_{X|Y}(x|y) p_Y(y)}.$$

With the conditional probability, we may fix $Y = y$ such that we

have a **conditional RV** $X|Y = y$.

$$p_{X|Y=y} = \frac{P(X = x, Y = y)}{P(Y = y)},$$

with **conditional expectation** $E[X|Y = y]$.

We may take Y as the random variable, to obtain the RVs **conditional expectation** $E[X|Y]$, and **conditional variance** $\text{Var}(X|Y)$.

Random Sum

For IID RVs $X_1, X_2, \dots$, and discrete RV N independent of X_i s, the random sum is defined

$$Y = \sum_{i=1}^N X_i.$$

By the laws of total expectation/variance, we have the results:

$$E[Y] = E(N\mu) = \mu E(N)$$

$$\text{Var}(Y) = E[N\sigma^2] + \text{Var}(N\mu)$$

$$= \sigma^2 E[N] + \mu^2 \text{Var}(N)$$

$$M_Y(t) = E[e^{t \sum_{i=1}^N X_i}] = E[E[e^{t \sum_{i=1}^N X_i} | N]]$$

$$= E[(M_X(t))^N] = E[e^{N \ln M_X(t)}]$$

$$= M_N(\ln[M_X(t)])$$

2 Markov Chains

2.1 Stochastic Processes

A **stochastic process** is a collection of indexed random variables $\{X(t, \omega) : t \in T\}$.

- Given an outcome ω , $X(\cdot, \omega) : T \rightarrow S$ is a sample path.

- Given an index t , $X(t, \cdot) : \Omega \rightarrow S$ is a distribution on S .

It is defined with a **state space** S that contains all possible states, and a **probability measure** P .

Classification

If T is countable, we have a **discrete-time** stochastic process. Otherwise, it is a **continuous-time** stochastic process.

If S is countable, we have a **discrete-state** stochastic process. Furthermore if S is finite, it is a **finite-state** stochastic process. Otherwise, it is a **real-state** stochastic process.

Specification

The probability measure for each outcome, of the joint distribution of $(X_{n_1}, X_{n_2}, \dots, X_{n_k})$ is too difficult to specify in general.

2.2 Markov Chain

A stochastic process satisfies the **Markovian property** if given current state X_n , the next state X_{n+1} is independent of all past states $X_{n-1}, \dots, X_0$.

$$P(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = P(X_{n+1} = j | X_n = i).$$

This $p_{ij}^{n,n+1}$ is known as the **(one-step) transition probability**.

$\{X_n : n \in T\}$ is called a **Markov Chain** if it is a **discrete-time discrete-state stochastic process** satisfying the Markovian property.

Specification

A Markov Chain can be **specified** (to find the joint distribution of X_n at any k time points) with

- The index set T

- The state space S

- The one-step transition probability $\forall i, j \in S, n \in T$

We define the **one-step transition probability matrix** $\mathbf{P}^{n,n+1}$ where each $\mathbf{P}_{ij}^{n,n+1} = P(X_{n+1} = j | X_n = i)$. The row vector yields the conditional distribution $X_{n+1} | X_n = i$.

Chapman-Kolmogorov Equations

For any $m, n \geq 0$, we have the equations

$$\mathbf{P}^{n,n+m+1} = \mathbf{P}^{n,n+1} \mathbf{P}^{n+1,n+m+1}$$

$$\mathbf{P}^{n,n+m+1} = \mathbf{P}^{n,n+m} \mathbf{P}^{n+m,n+m+1}.$$

$$p_{ij}^{n,n+m} = \sum_{k_1, k_2, \dots, k_{m-1}} p_{ik_1}^{n,n+1} \dots p_{k_{m-1}j}^{n+m-1,n+m}.$$

2.3 Stationary MC

With certain problems, we may further assume that $p_{ij}^{n,n+1}$ does not depend on n . We have the **stationary transition probability**

$$p_{ij}^{n,n+1} = p_{ij}, \quad i, j \in S, n \in T.$$

We only require $S, T, \mathbf{P}$ for the specification of a **stationary MC**.

$$\mathbf{P}^{(m+n)} = \mathbf{P}^{(m)} \mathbf{P}^{(n)} \quad \text{for stationary MCs.}$$

Finite-state stationary MCs can be represented with **graphs**, where nodes represent state, and edges represent possible transitions.

Random Walks

Suppose $\{\xi_n\}_{n \in \mathbb{N}^*}$ a sequence of IID random vectors with $\xi_i \sim p$. A stationary MC $\{X_n\}_{n \in \mathbb{N}}$, with $T = \mathbb{N}, S = \mathbb{Z}$ where

$$X_0 = x_0 \in \mathbb{Z}^n, \quad X_{n+1} = X_n + \xi_{n+1}, n \in \mathbb{N}.$$

is known as a **random walk**.

$$X_n = X_0 + \sum_{i=1}^n \xi_i, \quad (\xi_i + 1)/2 \sim \text{Bernoulli}(p).$$

If $X_0 = 0, (X_n + n)/2 \sim \text{Bin}(n, p)$.

Special Cases of (1-dim) random walks:

- Symmetric/Simple: $P(\xi_n = 1) = \frac{1}{2}$

- Biased: $P(\xi_n = 1) = p \neq \frac{1}{2}, \quad P(\xi_n = -1) = q$

- Lazy: $\xi_n \in \{-1, 1, 0\}, P(\xi_n = 0) = r > 0$

- with Reflecting Boundaries: for $a, b \in \mathbb{R}, a < b$,

$$P(X_{n+1} = a + 1 | X_n = a) = 1$$

$$P(X_{n+1} = b - 1 | X_n = b) = 1.$$

A random walk in $\mathbb{Z}^d$, where ξ_n is a step towards one of its $2d$ nearest neighbours, is known as a **d-dimensional random walk**.

Brownian Motion

Consider a random walk where in each step, the walker moves $\Delta x \rightarrow 0$ in $\Delta t \rightarrow 0$ time. This describes a continuous path. If $\Delta t = c\Delta x^2$ for some constant c , it is a **Brownian motion**.

2.4 Short-Term Analysis

For a MC with state space S , index set $\mathbb{N}$ and transition probability matrix $\mathbf{P}$, we have the following results:

- $P(X_1 = j | X_0 = i) = p_{ij}$

- Distribution of the **next state**: $(X_1 | X_0 = i) \sim \mathbf{P}_{i,\cdot}$.

- Distribution of **n-th state**: $(X_n | X_0 = i) \sim (\mathbf{P}^n)_{i,\cdot}$.

- Joint distribution of **two timepoints** $n \geq m \geq 0$

$$P(X_n = j, X_m = k | X_0 = i) = (\mathbf{P}^{n-m})_{kj} (\mathbf{P}^m)_{ik}$$

- Random initial state** $X_0 \sim \pi$

$$(X_n = j | X_0 \sim \pi) \sim \pi \mathbf{P}^{(n)}.$$

- Past state conditional** on current and initial state

$$P(X_m | X_n = j, X_0 = i) = \frac{(\mathbf{P}^{n-m})_{kj} (\mathbf{P}^m)_{ik}}{(\mathbf{P}^n)_{ij}}.$$

2.5 First Step Analysis

Definitions

For strategy-related questions involving the time index t , we first define a RV $T : \Omega \rightarrow \{0, 1, 2, \dots, \infty\}$ such that $\forall n \geq 0$,

$\{T = n\}$ depends solely on $X_1, X_2, \dots, X_n$,

$\{T \leq n\}$ depends solely on $X_1, X_2, \dots, X_n$.

T is a **stopping time**. Further, a state i is an **absorbing state** if

$$\forall j \in S - \{i\}, p_{ij} = 0$$

First Step Analysis

Using First Step Analysis, we may solve **problems related to stopping time** T , such as the state at time T , $E[T]$, functions of T .

Consider $a_i(T) = h(i, \dots, X_T | X_0 = i)$:

- Consider the first step, where

$$a_i(T) = \sum_{k \in S} h(i, k, X_2, \dots, X_T | X_0 = i, X_1 = k) P(X_1 = k | X_0 = i).$$

- Consider the probabilistic equivalent process $\{Y_n\}$.

$$h(i, k, X_2, \dots, X_T | X_0 = i, X_1 = k)$$

$$= g(h(k, X_1, \dots, X_T | X_0 = k)).$$

where g depends on the specified problem.

- Define $a_i(T) = h(i, \dots, X_T | X_0 = i)$ for each state $i \in S$, and express the equation as

$$a_i(T) = \sum_{k \in S} g(a_k(T)) P(X_1 = k | X_0 = i).$$

- Solving the system of equations of $a_i, \forall i \in S$, we obtain the desired result.

Strategy-Related Problems

To obtain $P(X_T = S_A | X_0 = S_i)$, we take the first step of the MC with the Chapman-Kolmogorov equations. Consider $Y_n = X_{n+1}$. $\{Y_n\}$ has the **same probabilistic structure** as $\{X_n\}$.

$$P(X_T = A | X_1 = i) = P(Y_{T-1} = 0 | Y_0 = i).$$

$T_Y := \min\{n : Y_n = S_A\}$. T_Y is a stopping time of $\{Y_n\}$.

$$P(Y_{T_Y} = 0 | Y_0 = i) = P(X_T = 0 | X_0 = i).$$

$$\therefore P(X_T = 0 | X_1 = 1) = P(Y_{T-1} = 0 | Y_0 = 1)$$

$$= P(Y_{T_Y} = 0 | Y_0 = 1) = P(X_T = 0 | X_0 = 1)$$

Defining $u_i = P(X_T = S_A | X_0 = S_i)$, we have a system of equations in $u_i, \forall i \in S$.

General Gambler's Ruin

Suppose a gambler who will stop with N , or 0 dollars in hand. In each game, they win with probability p , and lose with $q = 1 - p$. Beginning with $0 < k < N$ in hand, the **probability of stopping at 0** $u_k = P(X_T = 0 | X_0 = k)$ is

$$\begin{cases} u_k = 1 - \frac{k}{N} & p = q \\ u_k = 1 - \frac{1 - (q/p)^k}{1 - (q/p)^N} & p \neq q \end{cases}$$

The **expected number of games** played $v_k = E[T | X_0 = k]$ is

$$\begin{cases} v_k = k(N - k) & p = q \\ v_k = \frac{1}{p - q} \left[\frac{N(1 - (q/p)^k)}{1 - (q/p)^N} - k \right] & p \neq q \end{cases}$$

3 Long Run Performance

3.1 Classification of States

Communicating Classes

State j is **accessible from state** i , $i \rightarrow j$ if

$$\mathbf{P}_{ij}^m > 0 \text{ for some } m \geq 0.$$

If $i \rightarrow j \wedge j \rightarrow i$, they are **communicating**, $i \leftrightarrow j$.

Communication is an **equivalence relation**. We define $(S / \leftrightarrow)$ the set of communication classes, each denoted $\mathcal{C}$. It is **well-defined**.

Classes do not communicate. If $i \in \mathcal{C}_1, j \in \mathcal{C}_2$ such that $i \rightarrow j$, $\forall i' \in \mathcal{C}_1, j' \in \mathcal{C}_2 \quad i' \rightarrow j', \quad j \not\rightarrow i', \quad j' \not\rightarrow i'$.

A MC is **irreducible** if there is only one communication class.

Recurrence

For state i , the **first return probability** is the probability of having the n -th transition be the first return to i with initial state i

$$f_{ii}^{(0)} := 0.$$

$$f_{ii}^{(n)} = P(X_1 \neq i, \dots, X_{n-1} \neq i, X_n = i | X_0 = i).$$

The probability of **revisiting** i in the future is

$$f_{ii} = \sum_{n=0}^{\infty} f_{ii}^{(n)} \leq 1,$$

and the **return probability** to i from i at the n -th transition is

$$\mathbf{P}_{ii}^{(n)} = P(X_n = i | X_0 = i).$$

We have the **convolution**

$$P_{ii}^{(n)} = \sum_{k=1}^n f_{ii}^{(k)} P_{ii}^{(n-k)}.$$

The **expected number of revisits** to state i , $E[N_i | X_0 = i]$ is

$$E[N_i | X_0 = i] = \sum_{n=1}^{\infty} P_{ii}^{(n)} = \begin{cases} \infty & f_{ii} = 1 \\ \frac{f_{ii}}{1-f_{ii}} & f_{ii} < 1 \end{cases}$$

$$\begin{aligned} i \text{ is } \mathbf{recurrent/not recurrent} &\iff f_{ii} = 1 \\ &\iff E[N_i | X_0 = i] \text{ is infinite} \\ &\iff \sum_{n=1}^{\infty} P_{ii}^{(n)} \end{aligned}$$

Recurrence is a **class property**.

Periodicity

For a state i , we define the **period** of i

$$d(i) := \begin{cases} \gcd R_{ii} & R_{ii} \neq 0 \\ 0 & R_{ii} = 0 \end{cases}$$

where $R_{ij} := \{n \geq 1 : P_{ij}^{(n)} > 0\}$.

$$\begin{aligned} &\bullet \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, 0 < k < d(i) \\ &\quad \mathbf{P}_{ii}^{(n+d(i))} > 0, \quad \mathbf{P}_{ii}^{(n+d(i)+k)} = 0. \\ &\bullet \forall m \in R_{ji}, \text{ for the same } N, \forall n \geq N \\ &\quad \mathbf{P}_{ji}^{(m+nd(i))} > 0. \end{aligned}$$

Periodicity is a **class property**. A class with period 1 is **aperiodic**.

Regularity and Ergodicity

A MC is **regular** if $\exists k \in \mathbb{N}$ such that

$$\forall i, j \in S, \quad (\mathbf{P}^{(k)})_{ij} > 0.$$

$\mathbf{P}$ of a regular MC is a regular transition probability matrix.

3.2 Recurrent Irreducible MC

The **expected duration between revisits** is defined

$$m_i = E[R_i | X_0 = i] = \sum_{n=1}^{\infty} n f_{ii}^{(n)}.$$

If m_i is finite, the MC is **positive recurrent**. Otherwise, it is null recurrent. Given a fixed path, consider the **long-run proportion of visits**. It is given by $\lim_{n \rightarrow \infty} E\left[\frac{1}{n} \sum_{k=1}^n I(X_k = j | X_0 = i)\right]$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \mathbf{P}_{ij}^{(k)} = \frac{1}{m_j}.$$

If the MC is periodic: $\lim_{n \rightarrow \infty} \mathbf{P}_{jj}^{(nd)} = \frac{d}{m_j}$.

If the MC is aperiodic and null recurrent: $\lim_{n \rightarrow \infty} \mathbf{P}_{ij}^{(n)} = 0$.

Finite MC: Regular $\iff$ Ergodic $\iff$ Irreducible, Aperiodic.

Infinite MC: Ergodic $\iff$ Irreducible, Aperiodic, + Recurrent.

Ergodic MC

$\lim_{n \rightarrow \infty} \mathbf{P}_{ij}^{(n)}$ exists independent of initial state i .

$$\lim_{n \rightarrow \infty} \mathbf{P}_{ij}^{(n)} = \lim_{n \rightarrow \infty} \mathbf{P}_{jj}^{(n)} = \frac{1}{m_i} = \pi_j.$$

Since $\sum_{k=1}^N \pi_k = 1$, we define the **limiting distribution**

$$\pi := \langle \pi_1, \pi_2, \dots, \pi_N \rangle.$$

π is obtained from the solution to $\pi \mathbf{P} = \pi$, $\|\pi\|_1 = 1$

$$\begin{cases} \pi_j = \sum_{k=1}^N \pi_k \mathbf{P}_{kj} & j \in \{1, 2, \dots, N\} \\ \|\pi\|_1 = 1 \end{cases}$$

The regular transition probability matrix converges

$$\lim_{n \rightarrow \infty} \mathbf{P}^n = \begin{pmatrix} \pi \\ \pi \\ \vdots \\ \pi \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ \vdots \\ N \end{pmatrix} \begin{pmatrix} \pi_1 & \pi_2 & \dots & \pi_N \\ \pi_1 & \pi_2 & \dots & \pi_N \\ \vdots & \vdots & \ddots & \vdots \\ \pi_1 & \pi_2 & \dots & \pi_N \end{pmatrix}$$

That is, $X_n \sim \pi^{(0)} \mathbf{P}^{(n)} \rightarrow \pi$ for any initial distribution.

Given a large time point n , X_n is a **random variable**. Each path gives a realisation of X_n and $P(X_n = k) = \pi_j$.

Given a path, consider the **proportion of visits** over time.

$E\left[\frac{1}{n} \sum_{k=0}^{n-1} I(X_k = j | X_0 = i)\right] \rightarrow \pi_j$ as $n \rightarrow \infty$.

The **mean duration between visits** to j is given by $\frac{1}{m_j}$.

3.3 General MC Long Run Performance

1. Obtain the set of communicating classes $\mathcal{C}_k$
2. Set up a MC such that each recurrent class is denoted by a node
3. Define $T := \min\{n : X_n \in \text{recurrent classes}\}$
4. Using FSA, find $u_{k|i} = P(X_T \in \mathcal{C}_k | X_0 = i)$.
5. Within each recurrent class $\mathcal{C}_k$:

- Period > 1 : Long-run proportion of time in each state.
- Period $= 1$: If finite or infinite positive recurrent, find the corresponding limiting distribution of state $j \in \mathcal{C}_k$, denoted $\pi_{j|k}$. Otherwise, the probability at each state is 0.

6. For any initial state $X_0 = i$:

- If j is transient, $\pi_{j|i} = 0$
- If $j \in \mathcal{C}_k$, $\pi_{j|i} = u_{k|i} \pi_{j|k}$

7. With an initial distribution $X_0 \sim \pi^{(0)}$

$$\pi_{j|\pi^{(0)}} = \sum_{i \in S} \pi_{j|i} \pi_i^{(0)}.$$

3.4 Balance Equations

The **global balance equations** $\pi = \pi \mathbf{P}$ yields the limiting distribution. Equivalently, we have the N equations:

$$\pi_j = \sum_{k \in S} \pi_k \mathbf{P}_{kj}.$$

π exists even for non-ergodic MCs: $(X_{n+1} | X_n \sim \pi) \sim \pi$.

	Limiting Dist.	Stationary Dist.
Initial dist.	$X_0 \sim \pi^{(0)}$	$X_0 \sim \pi$
First step	$X_1 \sim \pi^{(0)} \mathbf{P}$	$X_1 \sim \pi \mathbf{P} = \pi$
$n \rightarrow \infty$	$X_n \sim \pi$	$X_n \sim \pi$
Calculation	$\pi = \pi \mathbf{P}$ or $\lim_{n \rightarrow \infty} \mathbf{P}^n$	$\pi = \pi \mathbf{P}$
Uniqueness	Yes	No
	$\pi_1 = \pi_2$	it's possible $\pi_1 \neq \pi_2$
MC	Ergodic	No requirement

A **unique** stationary distribution is the limiting distribution.

Consider a transition matrix $\mathbf{M}$ such that

$$\mathbf{M}_{ij} = P(X_{n+1} = j, X_n = i) = \pi_i \mathbf{P}_{ij},$$

The row sum equals the column sum for any $i, j \in S$. This can be obtained with the $\binom{n}{2}$ **local (detailed) balance equations**. Suppose there exists a distribution π such that

$$\pi_i \mathbf{P}_{ij} = \pi_j \mathbf{P}_{ji}.$$

Then matrix $\mathbf{M}$ is symmetric. π also satisfies the **global balance equations**, and is easier to compute.

4 Applications of MC

4.1 PageRank

Given a series of webpages S , each having k_i hyperlinks to other webpages, consider a random walk on the stationary MC:

- State Space S , Time index $\mathbb{N}$

$$P_{ij} = \begin{cases} \frac{1}{k_i} & \text{if there is a hyperlink } i \rightarrow j \\ 0 & \text{otherwise} \end{cases}$$

$$\pi^{(0)} = \left(\frac{1}{|S|}, \dots, \frac{1}{|S|} \right)$$

If the MC is nonregular, consider instead

$$\tilde{\mathbf{P}} = \lambda \mathbf{P} + (1 - \lambda) \pi_0 \mathbf{1}^T$$

$$\pi_{n+1} = \lambda \pi_n \mathbf{P} + (1 - \lambda) \pi_0.$$

for some damping factor $\lambda \in (0, 1)$. The limiting distribution π is used to rank in order of decreasing limiting probability.

4.2 Markov Chain Monte Carlo

Given a target distribution $\mathbf{p} = c_n \mathbf{p}'$, we find a MC such that $\pi = \mathbf{p}$, using the **Metropolis-Hastings Algorithm**:

1. Initialise with any initial state X_0 , and an ergodic MC with the same sample space as $\mathbf{p}$ and transition probability matrix $\mathbf{Q}$
2. Using $\mathbf{Q}_{X_n}$, draw a proposal state Y .
3. Calculate the acceptance rate of the proposal state

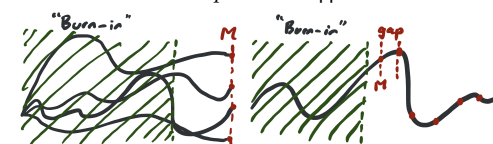
$$\alpha(X_n, Y) = \min \left(\frac{p'_Y}{p'_{X_n}} \cdot \frac{Q_{Y, X_n}}{Q_{X_n, Y}}, 1 \right).$$

4. If the proposal state is accepted, $X_{n+1} = Y$.

On a **continuous** distribution $f = c_n f'$, the state space S is a continuous set, the transition probability is the **density function** $g_X(y) = g(y|x)$ and the **acceptance rate** is the analogue

$$\min \left(\frac{f'(Y)}{f'(X_n)} \cdot \frac{g_Y(X_n)}{g_{X_n}(Y)}, 1 \right)$$

For sufficiently large M and gap , we can simulate sampling from p from the MC to simulate p with two approaches.



4.3 Branching Process

A **branching process** is a stochastic process $\{X_n\}_{n=0}^{\infty}$ where, for some IID RVs $\xi_i^{(j)}$ over the sample space $\mathbb{N}$,

$$X_{n+1} = \sum_{i=0}^{X_n} \xi_i^{(n)}.$$

Properties of Branching Processes: A **stationary MC** of known transition probability function, **aperiodic** with two state classes: $\{0\}$ and $\mathbb{N}^*$.

$$\phi_{X_{n+1}|X_n}(t) = \phi_{\xi}^{X_n}(t)$$

$$\phi_{X_{n+1}}(t) = E\left[\phi_{\xi}^{X_n}(t)\right] = \phi_{X_n}(\phi_{\xi}(t))$$

$$\text{If } X_0 = k, \phi_{X_n}(t) = \left[\phi_{\xi}^{(n)}(t)\right]^k.$$

With the known **moments of ξ** ,

$$E[X_n] = \mu E[X_{n-1}] = \mu^n.$$

$$Var(X_n) = \mu^{n-1} \sigma^2 \times \begin{cases} \frac{1-\mu^n}{1-\mu} & \mu \neq 1 \\ n & \mu = 1 \end{cases}.$$

5 Poisson Process

For absorbing state 0, we define the stopping time and denote

$$T = \min\{n : X_n = 0\}, \quad u_n^{(k)} := P(T \leq n | X_0 = k)$$

Since $\{T \leq n\} = \cap_{i=1}^n \{T \leq i\} \implies u_n^{(k)} = u_n^{(k)}$,

$$u_n = \sum_{l \in S} P(\xi = l) u_{n-1}^l = \phi_{\xi}(u_{n-1}),$$

and $u_1 = P(\xi = 0)$, we can find the **extinction probability before generation n** $u_n^{(k)}$ by

$$u_n^{(k)} = u_n^k = \left[\phi_{\xi}^{(n-1)}(P(\xi = 0)) \right]^k.$$

The eventual **extinction probability** exists as it is a non-decreasing probability wrt. n . $\phi_{\xi}(0) = P(\xi = 0)$, $\phi_{\xi}(1) = 1$, and it is a **convex increasing function** wrt x , we have the classes of solutions:

- $P(\xi = 0) = 0 \implies u_{\infty} = 0$
 - $P(\xi = 0) > 0, E[\xi] < 1 \implies u_{\infty} = 1$. **Subcritical**.
 - $P(\xi = 0) > 0, E[\xi] = 1 \implies u_{\infty} = 1$. **Critical**.
 - $P(\xi = 0) > 0, E[\xi] > 1 \implies u_{\infty} < 1$. **Supercritical**.
- u_{∞} is the solution to $x = \phi_{\xi}(x)$.

5 Poisson Process

For independent events over $[0, t_0]$ decomposed into $n \rightarrow \infty$ intervals of length $h = \frac{t_0}{n}$, $N((kh, kh+h]) \sim \text{Bern}(p_n)$ and $N([0, t_0]) \sim \text{Bin}(n, p_n)$. If $p_n \in o(h)$, $np_n \rightarrow \lambda$,

$$\begin{aligned} P(X = x) &= \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \\ &= \frac{n! / (n-x)!}{n^x} \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^{n-x} \rightarrow \frac{\lambda^x}{x!} e^{-\lambda} \end{aligned}$$

We have the **Poisson Distribution** $X \sim \text{Pois}(\lambda)$ with $\lambda > 0$,

$$P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}, \quad x \in \mathbb{N}.$$

$$E[X] = Var(X) = \lambda, \quad M_X(t) = e^{\lambda(e^t - 1)}.$$

A **Poisson process of rate/intensity** $\lambda > 0$ is a continuous-time, discrete-state stochastic process $\{X(t) | t \geq 0\}$ where

- $X(t_1) - X(t_0), X(t_2) - X(t_1), \dots, X(t_n) - X(t_{n-1})$ are independent RVs, for any $t_0 = 0 < t_1 < \dots < t_n$
- $X(s+t) - X(s) \sim \text{Pois}(\lambda t)$, $s \geq 0, t > 0$
- $X(0) = 0$

Alternatively, by the **law of rare events**, $N((s, t])$ is a **Poisson process** of intensity $\lambda > 0$ if

- $N((t_0, t_1]), \dots, N((t_{n-1}, t_n])$
- $\exists \lambda, \quad P(N((t, t+h]) \geq 1) = \lambda h + o(h), \quad h \rightarrow 0$
- $\exists \lambda, \quad P(N((t, t+h]) \geq 2) = o(h), \quad h \rightarrow 0$

This allows generalisation: $\lambda \rightarrow \lambda(t)$ yields a **non-homogeneous Poisson process**, $h \rightarrow Vol(A)$ a **Poisson point process**.

For a Poisson process $X(t)$ of rate λ , the **waiting time** of the n -th event $W_n \sim \text{Exp}(\lambda) = \Gamma(n, \lambda)$. Further, the **sojourn times** $S_n = (W_{n+1} - W_n) \sim \text{Exp}(\lambda)$.

This gives the alternative definition: the counting process with waiting times $W_n = S_0 + S_1 + \dots + S_{n-1}$ is a **Poisson process** of rate λ if

$$\{S_n : n \geq 0\} \text{ are IID RVs, } S \sim \text{Exp}(\lambda)$$

With the memoryless property, given that $X(t) = n$, each of the events occur at any $t \in [0, t_0]$ with equal probability. That is, The **conditional distribution** of $W_1, W_2, \dots, W_n$, given $X(t) = n$, is the same as the **order statistics** corresponding to n IID uniform RVs over $(0, t)$.

$$f_{W_k}(x) = \frac{n!}{(n-k)!(k-1)!} \frac{1}{t} \left(\frac{x}{t}\right)^{k-1} \left(\frac{t-x}{t}\right)^{n-k}$$